

Marwan Abdelaal

Backend Engineer · Full-Stack Developer · AI Engineering Intern

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PROFILE

3rd-year Software Engineering student at GIU (GPA: 2.4 — Very Good, German Scale) with a Minor in IT Security. I build backend systems that work in the real world — production APIs, durable schedulers, idempotent publishing pipelines, and AI-integrated platforms. Currently interning at FlyRank AI where I shipped a multi-platform social campaign publisher with reliability engineering at its core. I led a team of 10 to ship an AI-powered MERN career platform, and hold 20 Anthropic Academy certifications in AI fluency and Claude tooling. I take ownership, ship things, and learn fast.

SKILLS

Backend:	Node.js · Express.js · REST APIs · JWT · Middleware · Supabase Auth · Swagger UI · BullMQ
Languages:	JavaScript · Python · Java · C# · SQL · HTML & CSS
Databases:	PostgreSQL · MongoDB · SQLite · SQL Server · Redis · Schema Design
AI & ML:	Hugging Face · Ollama · NLP · Prompt Engineering · Embeddings · Zero-Shot Classification
Security:	AES-GCM · HMAC · 3DES · RSA · Diffie-Hellman · OAuth · Webhook Verification
Reliability:	Idempotency · Durable Scheduling · Rate Limit Handling · Retry Logic · Exponential Backoff
DevOps:	Docker · Docker Compose · Git · GitHub · GitLab · CI/CD
Frontend:	React.js (Vite) · Context API · React Router · Axios
Data:	Cheerio · Zod · pandas · scikit-learn · Jupyter Notebook

EXPERIENCE

Backend AI Engineering Intern FlyRank AI	Jun 2026 — Present Remote
— Shipped a multi-platform social campaign publisher as capstone — idempotent publishing, durable BullMQ/Redis scheduler, HMAC-verified webhooks, AES-GCM encrypted OAuth tokens, and rate-limit-aware retry logic against a fake social platform server.	
— Built a production REST API with 12 endpoints, SQLite persistence, parameterized queries for SQL injection prevention, and 23 automated tests at 100% pass rate — then migrated to PostgreSQL in Docker using a clean repository pattern.	
— Implemented JWT authentication via Supabase Auth with reusable middleware and full Swagger UI documentation, and built a web scraping pipeline with Zod validation, caching (45s to 1s), and Ollama AI enrichment.	
— Completed 20 Anthropic Academy certifications spanning AI fluency, Claude API, MCP, agent skills, Amazon Bedrock, and Google Cloud.	

Part-Time Teaching Assistant ICTeam El Komy — IGCSE ICT	Jun 2024 — May 2026 Cairo, Egypt
— Guided 20+ students through technical concepts and lab sessions, troubleshooting issues in real time and adapting content delivery based on continuous feedback — strengthening communication and leadership under pressure.	

Event Volunteer & Usher Cairo International Film Festival & GIU Events	2024 — Present Cairo, Egypt
— Coordinated operations across 3+ large-scale public events, managing cross-functional volunteer teams and ensuring smooth execution under high-pressure conditions.	

PROJECTS

Multi-Platform Social Campaign Publisher — FlyRank Capstone
Node.js · Express.js · BullMQ · Redis · PostgreSQL · Docker · Sharp · AES-GCM · HMAC github.com/MoscowAbdelaal/flyrank-capstone-social-studio
— Converts a single blog post into a full social campaign — platform-correct image variants (Instagram 1080x1080, X 1600x900), platform-tailored captions from reusable prompt fragments, durable scheduling, and webhook-verified delivery tracking.
— Reliability at the core: idempotent publishing (duplicate requests yield one post), 429/Retry-After backoff, crash-safe BullMQ worker, HMAC signature-verified webhooks rejecting forgeries with 400, and AES-GCM encrypted tokens with random IV.
GIU Nexus — AI-Powered Career & Talent Platform
React (Vite) · Node.js · Express.js · MongoDB · Hugging Face API · JWT · GitLab github.com/MoscowAbdelaal/GIU-Nexus

— Led a 10-person team to build a full-stack MERN platform connecting university students with jobs — Hugging Face NLP for skill extraction and zero-shot job classification, cosine similarity recommendations, and role-based dashboards for job seekers, recruiters, and admins.

Production CRUD API — Auth, Testing & Docker Migration

Node.js · Express.js · SQLite · PostgreSQL · Docker · Supabase Auth · JWT · Swagger UI | github.com/MoscowAbdelaal/crud-api

— 12-endpoint REST API with full CRUD, 23 automated tests at 100% pass rate, JWT auth via Supabase, parameterized queries, Swagger docs, and a clean repository-pattern migration from SQLite to PostgreSQL in Docker.

Human Resource Management System (HRMS)

Microsoft SQL Server · Node.js · HTML · CSS · SQL Stored Procedures

— Designed full EERD and relational schema from scratch, implemented optimized SQL stored procedures as the backend layer, and delivered a multi-role system for payroll, attendance, leave, hiring, and performance reviews.

Cryptography Suite — 3DES & Diffie-Hellman/RSA

Python · Jupyter Notebook

— Implemented Triple DES from scratch using EDE structure across 16-round blocks, and built a secure communication system combining Diffie-Hellman key exchange with RSA digital signatures — simulating HTTPS-like secure transmission.

Clash of Clans: Heroes — 2-Player Strategy Game

Java · JavaFX · Eclipse IDE · OOP · CSV Data Loading | github.com/MoscowAbdelaal/Clash-of-clans-Legends

— Independently built a complete 2-player turn-based strategy game across 3 milestones — full OOP architecture with abstract classes, interfaces, inheritance, custom exception handling, and a JavaFX frontend with real-time state updates and victory detection.

CERTIFICATIONS — ANTHROPIC ACADEMY (2026)

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| — AI Fluency: Framework & Foundations | — Claude 101 |
| — Claude Code 101 | — Building with the Claude API |
| — Introduction to Model Context Protocol | — Model Context Protocol: Advanced Topics |
| — Claude with Amazon Bedrock | — Claude on Google Cloud |

20 certifications total — full list available at [linkedin.com/in/marwan-abdelaal](https://www.linkedin.com/in/marwan-abdelaal)

EDUCATION

Bachelor of Software Engineering — Informatics & Computer Science
German International University (GIU)

Sep 2024 — Present | Cairo, Egypt

GPA: 2.4 / Very Good (German Scale) · Minor: IT Security · Expected Graduation: 2028 · Coursework: DSA, OOP, Database Systems, Data Science, Information Security, Distributed Systems, Media Informatics

LANGUAGES

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| — Arabic Native | — English Fluent |
| — German Beginner | — French Beginner |